📘 Chapter 8 - Matrices and Determinants

8.1 Matrices (Matrizen)

◆ Definition

A **matrix (Matrix)** is a rectangular arrangement of real numbers with m rows (Zeilen) and n columns (Spalten):

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}$$

- The element a_{ij} is in the *i-th row* and *j-th column* (i-te Zeile, j-te Spalte).

◆ Special Matrices (spezielle Matrizen)

1. **Column vector (Spaltenvektor):**

$$A = \begin{pmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{pmatrix}$$

2. **Row vector (Zeilenvektor):**

$$A = (a_{11}, a_{12}, \dots, a_{1n})$$

3. **Square matrix (quadratische Matrix):** $m = n$
4. **Upper triangular matrix (obere Dreiecksmatrix):**

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ 0 & a_{22} & a_{23} & \dots & a_{2n} \\ 0 & 0 & a_{33} & \dots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & a_{nn} \end{pmatrix}$$

   (analogous: lower triangular matrix – *untere Dreiecksmatrix*)

◆ Matrix-Vector Product (Matrix-Vektor-Produkt)

Definition:
The product of an (m, n)-matrix A with a vector $\vec{b}$ having n entries is a new vector $\vec{c}$ with m entries:

$$\vec{c} = A \cdot \vec{b} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{pmatrix}$$

$$\Rightarrow c_i = a_{i1}b_1 + a_{i2}b_2 + \cdots + a_{in}b_n$$

Each entry of $\vec{c}$ is the **scalar product (Skalarprodukt)** of a row of A with $\vec{b}$.

◆ Matrix Product (Matrixprodukt)

If A is a (p, m)-matrix and B an (m, n)-matrix, the product $C = A \cdot B$ is defined as:

$$c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{im}b_{mj}$$

- C is a (p, n)-matrix.
- The number of columns of A must equal the number of rows of B.
- In general: $A \cdot B \neq B \cdot A$ (**not commutative – nicht kommutativ**).

8.2 Determinants (Determinanten)

◆ 2×2 Determinant

For a $(2, 2)$-matrix:

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

the determinant is:

$$\det(A) = a_{11}a_{22} - a_{12}a_{21}$$

Rule: **Main diagonal minus secondary diagonal**
(Diagonale minus Nebendiagonale)

◆ Properties of the Determinant (Eigenschaften der Determinante)

1. Swapping rows or columns changes the **sign (Vorzeichen)**.
2. Multiplying a row/column by a scalar multiplies the determinant by that scalar.
3. If two rows/columns are identical or proportional → $\det(A) = 0$.
4. Adding a multiple of one row/column to another does **not** change $\det(A)$.
5. $\det(A) = 0$ if A has a zero row (Nullzeile) or column (Nullspalte).

◆ 3×3 Determinant — Sarrus' Rule (Regel von Sarrus)

$$\det(A) = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

can be calculated by:

$$\det(A) = (a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32}) - (a_{13}a_{22}a_{31} + a_{12}a_{21}a_{33} + a_{11}a_{23}a_{32})$$

Mnemonic: **"Main diagonals minus secondary diagonals"**
(Diagonalen minus Nebendiagonalen)

🔹 Determinant and Vector Operations (Determinanten und Vektoroperationen)

Cross product (Vektorprodukt):

$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{e}_x & \vec{e}_y & \vec{e}_z \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix}$$

Scalar triple product (Spatprodukt):

$$[\vec{a}\vec{b}\vec{c}] = \begin{vmatrix} a_x & b_x & c_x \\ a_y & b_y & c_y \\ a_z & b_z & c_z \end{vmatrix}$$

🔹 Laplace Expansion (Entwicklungssatz von Laplace)

For any $n \times n$ determinant D:

$$D = \sum_{j=1}^{n} (-1)^{i+j} a_{ij} D_{ij}$$

where D_{ij} is the **minor determinant (Nebendeterminante)** obtained by removing row i and column j.

The **signs follow a checkerboard pattern (Schachbrettmuster):**

$$\begin{vmatrix} + & - & + & \dots \\ - & + & - & \dots \\ + & - & + & \dots \end{vmatrix}$$

🔹 Practical Determinant Computation (Determinantenberechnung in der Praxis)

- For 2×2 or 3×3: use direct formula or Sarrus' rule
- For larger matrices: expand by rows or columns with many zeros
- Computationally efficient form: **Triangular form (Dreiecksform)**

If A is in upper triangular form:

$$\det(A) = a_{11} \cdot a_{22} \cdot a_{33} \cdots a_{nn}$$

🔹 Inverse Matrix (Inverse Matrix)

- $\det(A) = 0$ → **singular (singulär)** → not invertible
- $\det(A) \neq 0$ → **regular (regulär)** → invertible

If $\vec{y} = A \cdot \vec{x}$, then:

$$\vec{x} = A^{-1} \cdot \vec{y}$$

🔹 Formula for Inverse Matrix (Formel für die inverse Matrix)

$$A^{-1} = \frac{1}{\det(A)} \begin{pmatrix} D_{11} & -D_{21} & D_{31} & \cdots \\ -D_{12} & D_{22} & -D_{32} & \cdots \\ \vdots & \vdots & \ddots & \vdots \\ (-1)^{n+1} D_{1n} & (-1)^{n+2} D_{2n} & \cdots & (-1)^{n+n} D_{nn} \end{pmatrix}$$

- D_{ij} are **minor determinants (Nebendeterminanten)**.
- Note: In A^{-1}, element D_{ji} (transposed position) appears!
- The sign pattern follows the **checkerboard (Schachbrettmuster)** rule.

✅ Summary Table (Überblick)

Concept	German Term	Formula / Definition
Matrix	Matrix	rectangular array of numbers
Element a_{ij}	Matrixelement	entry in row i, column j
Determinant	Determinante	$\det(A)$
2×2 determinant	2×2-Determinante	$a_{11}a_{22} - a_{12}a_{21}$
3×3 determinant	3×3-Determinante	Rule of Sarrus
Minor	Nebendeterminante	determinant of submatrix
Laplace expansion	Entwicklungssatz von Laplace	$D = \sum(-1)^{i+j} a_{ij} D_{ij}$
Inverse matrix	Inverse Matrix	$A^{-1} = \frac{1}{\det(A)} \mathrm{Adj}(A)$
Singular matrix	singuläre Matrix	$\det(A) = 0$
Regular matrix	reguläre Matrix	$\det(A) \neq 0$
